Momentum in two dimensions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Conserve momentum separately along perpendicular axes.

SUCCESS CRITERIA - I can...

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “system” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

system: the object or objects chosen for analysis

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

Initial momentum is 0.3 kg·m/s and measured final momentum is 0.29 kg·m/s.
Find percent deviation.
Explain why this is not itself a measured uncertainty.

Word help: momentum = mass times velocity, including direction

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

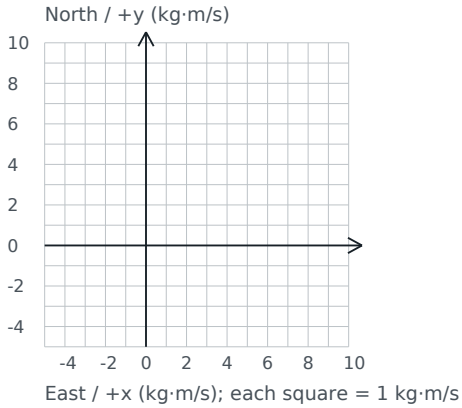
02 Worked example: explain the model

Reference: $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$. Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

Worked reasoning: Two objects have momenta (3, 0) and (0, 4) kg·m/s. Their total is (3, 4), magnitude 5 kg·m/s. Add components before finding the magnitude.

Your turn: Draw a tip-to -tail momentum diagram and label each axis with units.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

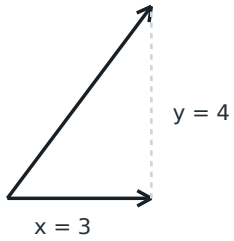
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two incoming arrows point east and north.
Sketch the direction of their vector sum. Can their ordinary lengths be added for the diagonal magnitude?

Word help: magnitude = size of a quantity without its direction

Sentence starter: The … in my model represents … . Its label shows … .



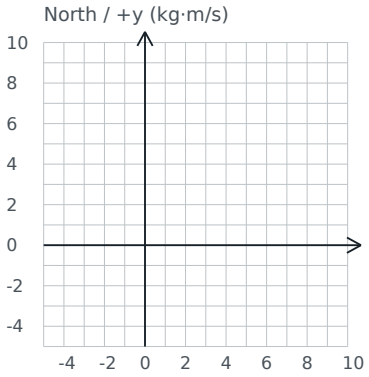
The arrows represent horizontal and vertical components; use their directions to describe the sum.

Marzano check - circle: 1 2 3

04 Apply the physics

A 2 kg cart at 3 m/s east sticks to a 1 kg cart at 4 m/s north.
Find final velocity components, speed and angle north of east.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Why must both momentum component sums use the same system and same time interval?

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

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Continue down each column.

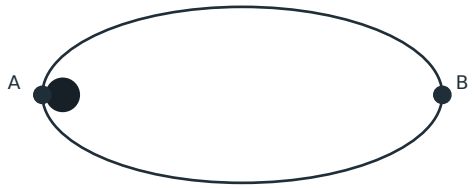
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Two ideal Sun-orbits have $a_B = 4\ a_A$.
Find T_B/T_A.
Explain why the same normalized Sun-orbit equation is not automatically valid around another body.
Label the orbital features used in your explanation.

Word help: orbit = a path around another body under gravity

Sentence starter: My design shows … . It meets the requirement by … .



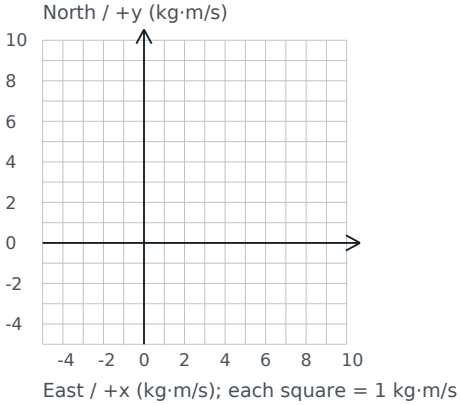
Marzano check - circle: 1 2 3

07 Exit check

Initial total momentum is (10, 0) kg·m/s. One fragment leaves with (4, 3).
Find the other momentum vector and its magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



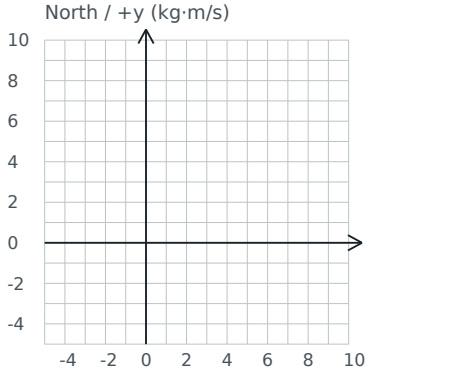
Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two pucks stick on an ideal air table. One has mass 1 kg and moves east at 4 m/s. The other has mass 3 kg and moves north at 2 m/s.
Draw the momentum sum.
Find the joined pair's velocity components, speed and direction.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3